Joel Smith

Machine Learning Systems Engineer

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Education

University of Liverpool Sept 2024 – Sept 2025
MSc in Data Science & Artificial Intelligence, Distinction

University of Chester Sept 2020 – July 2023
BSc in Economics, First Class (University Prize for Economics)

Selected Projects

fast-vllm | *Cold-start Optimisation for vLLM Inference* GitHub — Writeup

- CRIU checkpoint/restore of a warmed vLLM process, bypassing the cold-start sequence; startup reduced from 117s to 5.8s.
- Parallel double-buffered weight loader using multi-threaded preadv to build NVMe queue depth overlapped with h2d transfer, replacing vLLM's per-tensor reload path; 6GB reload time reduced from 8s to 2.4s (NVMe floor 1.6s).

vLLM | *Open-Source Contributions*

- Merged architecture-guard fix enabling per-tensor FP8 CUTLASS kernels on Blackwell (SM 12.1); contributing fixes while working through the v1 engine internals.

Alchemy Runtime | *TensorRT Inference Server* GitHub

- C++ inference server with TensorRT backend; dynamic batch queue (single drain thread, max-batch/max-wait window) with async request/response via future/promise behind a thread-pooled HTTP layer (httplib).
- On-device DDIM denoising loop minimising host-device transfers; custom CUDA kernel parallelising per-element posterior mean computation and stochastic sampling across the batch.
- Load-tested with wrk: 17 img/s peak throughput, p99 686ms at batch size 8 (RTX 5060 Ti).

Alchemy Lab | *Diffusion Model Research Framework* GitHub

- Modular diffusion training framework built from scratch: config-driven (Hydra/OmegaConf) sweeps and ablations, checkpoint/resume for interruptible runs, seed control for reproducibility, EMA for improved sample quality, instrumented with NVTX/Nsight for bottleneck analysis.
- Validated end-to-end with DDPM on CelebA-HQ 256 via PyTorch DDP across 4× RTX 4090 GPUs.

Experience

Artificial Intelligence Engineer April 2026 – Present
Currys *London, UK*

- Built tabular Q&A functionality for the company-wide LLM chatbot, enabling natural language queries over structured data on a Databricks stack.
- Migrated internal chatbot codebase to Azure DevOps and implemented CI/CD pipelines for build, test, and deployment.

Technical Skills

Inference & Systems: vLLM, TensorRT, CRIU, CUDA, Nsight Systems
Machine Learning: PyTorch, DDP, Hydra/OmegaConf
Languages: Python, C++, SQL
Infrastructure: Docker, Kubernetes, Azure DevOps, Databricks, AWS, Git, Linux
LLM & Retrieval: pgvector, ChromaDB

Technical Writing

Generative Modelling Blog

- In-depth articles covering generative model theory (GMM, VAE, diffusion) with full mathematical derivations.
- Supporting implementations in NumPy and PyTorch accompanying each article.